

Cloud Based AI Platform Comparison: Amazon Rekognition and Google AI Studio

Matthew Kolonich

Arin 320

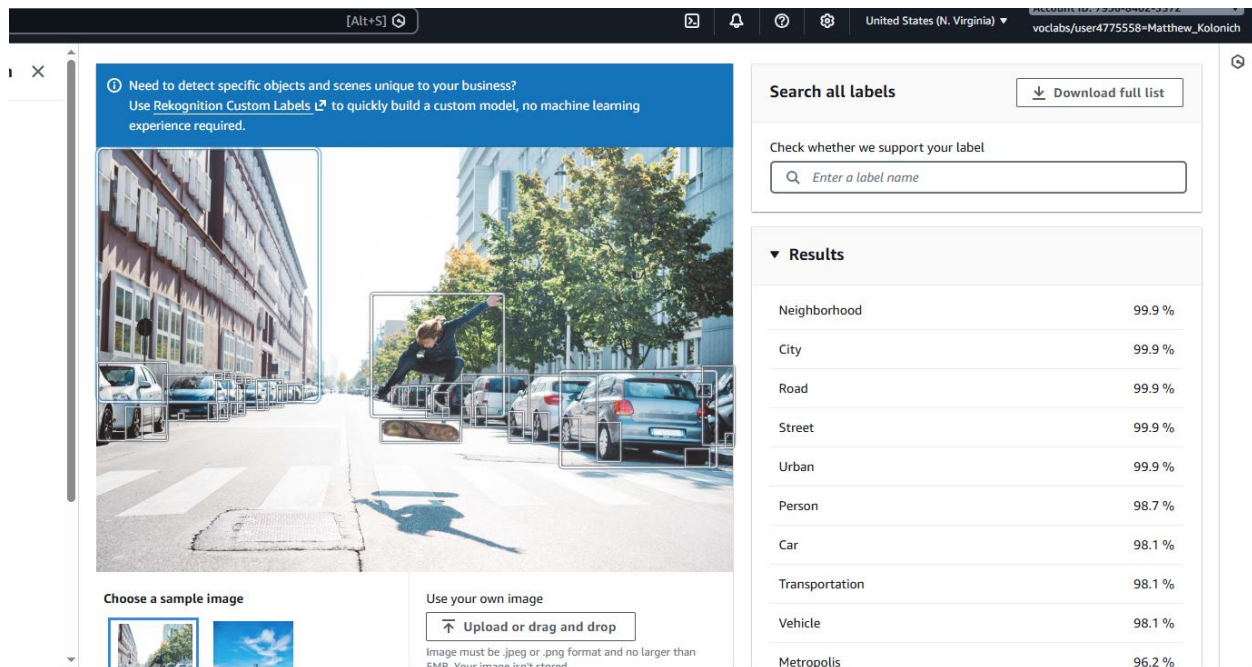
Professor Kirksey

Introduction

For this assignment, I compared Amazon Rekognition and Google AI Studio. I wanted to evaluate how different cloud based AI platforms support computer vision and multimodal tasks under real world constraints. I used these platforms because they represent two contrasting approaches to applied AI. Amazon Rekognition is designed as a production oriented service that emphasizes structured detection and confidence scoring. Google AI Studio in turn, prioritizes generative creativity and flexible multimodal reasoning. I evaluated both platforms through hands on testing. This allowed me to assess not only feature availability, but also how design choices influence reliability, interpretability, and suitability for domain specific use cases such as healthcare and law enforcement.

Amazon Rekognition Exploration and Testing

I explored Amazon Rekognition through the AWS Learner Lab environment, focusing on label detection. I also experimented with the celebrity recognition, face comparison, and image properties.



The screenshot displays the Amazon Rekognition console interface. On the left, a sample image of a city street is shown with bounding boxes around various objects. A blue banner at the top of the image area reads: "Need to detect specific objects and scenes unique to your business? Use Rekognition Custom Labels to quickly build a custom model, no machine learning experience required." Below the image, there are options to "Choose a sample image" (with two small thumbnails) or "Use your own image" (with an "Upload or drag and drop" button and a note: "Image must be .jpeg or .png format and no larger than 5MB. Your image isn't stored"). On the right side of the console, there is a "Search all labels" section with a "Download full list" button and a search input field labeled "Enter a label name". Below this is a "Results" section showing a list of detected labels and their confidence scores.

Label	Confidence Score
Neighborhood	99.9 %
City	99.9 %
Road	99.9 %
Street	99.9 %
Urban	99.9 %
Person	98.7 %
Car	98.1 %
Transportation	98.1 %
Vehicle	98.1 %
Metropolis	96.2 %

In the first sample image, Rekognition accurately labeled cars, a person, and wheels. The confidence scores appeared reasonable and aligned with visual clarity. However, contextual elements such as trees, windows, and the road were not labeled. This suggests that Rekognition prioritizes highly noticeable foreground objects instead of a full scene understanding. From a technical perspective, this behavior is consistent with object detection models optimized for precision rather than contextual completeness. In regulated domains, this design choice may limit situational interpretation.

Need to detect specific objects and scenes unique to your business?

Use [Rekognition Custom Labels](#) to quickly build a custom model, no machine learning experience required.



Choose a sample image



Use your own image

Upload or drag and drop

Image must be .jpeg or .png format and no larger than 5MB. Your image isn't stored.

Search all labels [Download full list](#)

Check whether we support your label

Enter a label name

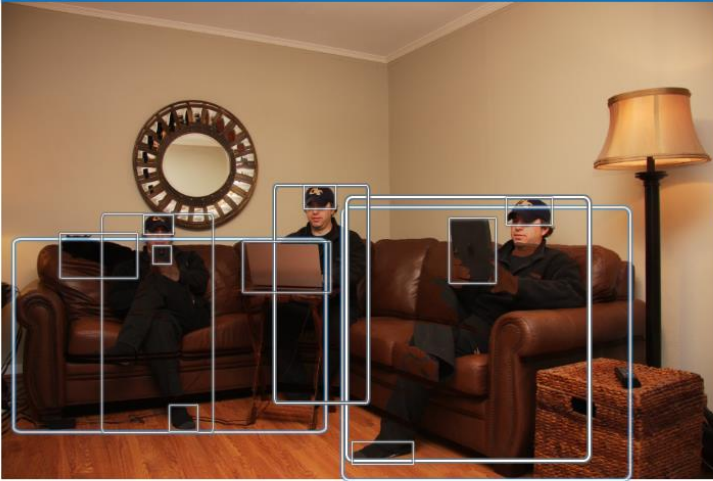
Results

Architecture	100 %
Building	100 %
City	100 %
Cityscape	100 %
Urban	100 %
Metropolis	99.8 %
Office Building	99.8 %
Water	99.1 %
Waterfront	99.1 %
High Rise	97.4 %

The second image produced only two labels, skyscraper and boat. The skyscraper confidence score of approximately 55 percent stood out as unusually low. It is pretty clear that there is a labeled skyscraper as well as others around the labeled skyscraper. This raised concerns about how confidence thresholds are calibrated and how easily users might misinterpret confidence values as measures of certainty rather than probabilistic estimates. In law enforcement or healthcare contexts, such misinterpretation could lead to over reliance or inappropriate dismissal of AI outputs.

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Choose a sample image



Use your own image

Upload or drag and drop

Search all labels [Download full list](#)

Check whether we support your label

Enter a label name

Results

Architecture	100 %
Building	100 %
Couch	100 %
Furniture	100 %
Indoors	100 %
Living Room	100 %

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In the third image, Amazon Rekognition demonstrated stronger overall performance by correctly identifying the setting as a living room. It was also able to detect multiple people within the scene. This suggests that the system performs more reliably when environments and human presence align with common visual patterns in its training data. However, technical limitations were still evident. The system mislabeled a tablet as a mobile phone. This indicates difficulty in distinguishing between visually similar consumer devices when contextual signals are subtle. In addition, several clearly visible objects, including a mirror, a lamp, and a control device, were not labeled at all. These omissions suggest that Rekognition prioritizes certain object categories and confidence thresholds over complete scene coverage. While this approach may reduce false positives, it also limits contextual understanding. This could be problematic in applications where missing environmental details affect interpretation or downstream decisions.

Need to detect specific objects and scenes unique to your business? Use [Rekognition Custom Labels](#) to quickly build a custom model, no machine learning experience required.



Choose a sample image

Use your own image

Upload or drag and drop

Search all labels

Download full list

Check whether we support your label

Glasses

Results

Face	99.9 %
Head	99.9 %
Person	99.9 %
Photography	99.9 %
Portrait	99.9 %
Accessories	99.9 %

Show more

Request

In the fourth image, which featured myself standing in front of a painting. Amazon Rekognition labeled only the painting and objects within the artwork. It while failed to detect or label me as a person. Technically, this outcome suggests that the system either did not reach its confidence threshold for face or person detection or deprioritized those features in favor of visually prominent elements within the painting. This result raises ethical considerations related to how automated systems interpret and abstract creative works. By focusing on the painting rather than the human subject, the system reduced a piece of artwork to a set of detectable objects without context. This raises questions about how visual art is categorized, reused, or interpreted in automated pipelines. This image highlights the issue of selective visibility in AI systems. What the system chose to recognize, and what it ignored, shapes how reality is represented. This in turn reinforces the importance of understanding not just AI accuracy, but also its blind spots.

In addition to the label detection, I explored three additional Amazon Rekognition demos: celebrity recognition, face comparison, and image properties. The celebrity recognition demo successfully identified well known public figures in sample images. It returned confidence scores along with metadata, demonstrating how the system matches detected faces against a curated reference

dataset. The face comparison demo allowed me to upload two images and evaluate whether they contained the same individual. This feature performed reliably when images were clear and well aligned. This reinforced its potential usefulness for identity verification workflows, while also highlighting the importance of image quality and lighting conditions. Finally, the image properties demo analyzed technical attributes such as brightness, sharpness, and contrast. While these outputs did not directly contribute to semantic understanding, they offered useful insights for preprocessing and quality control. This would be particularly useful in pipelines where image clarity affects downstream detection accuracy. Together, these demos illustrated how Rekognition supports both high level recognition tasks and lower level image diagnostics within a unified service.

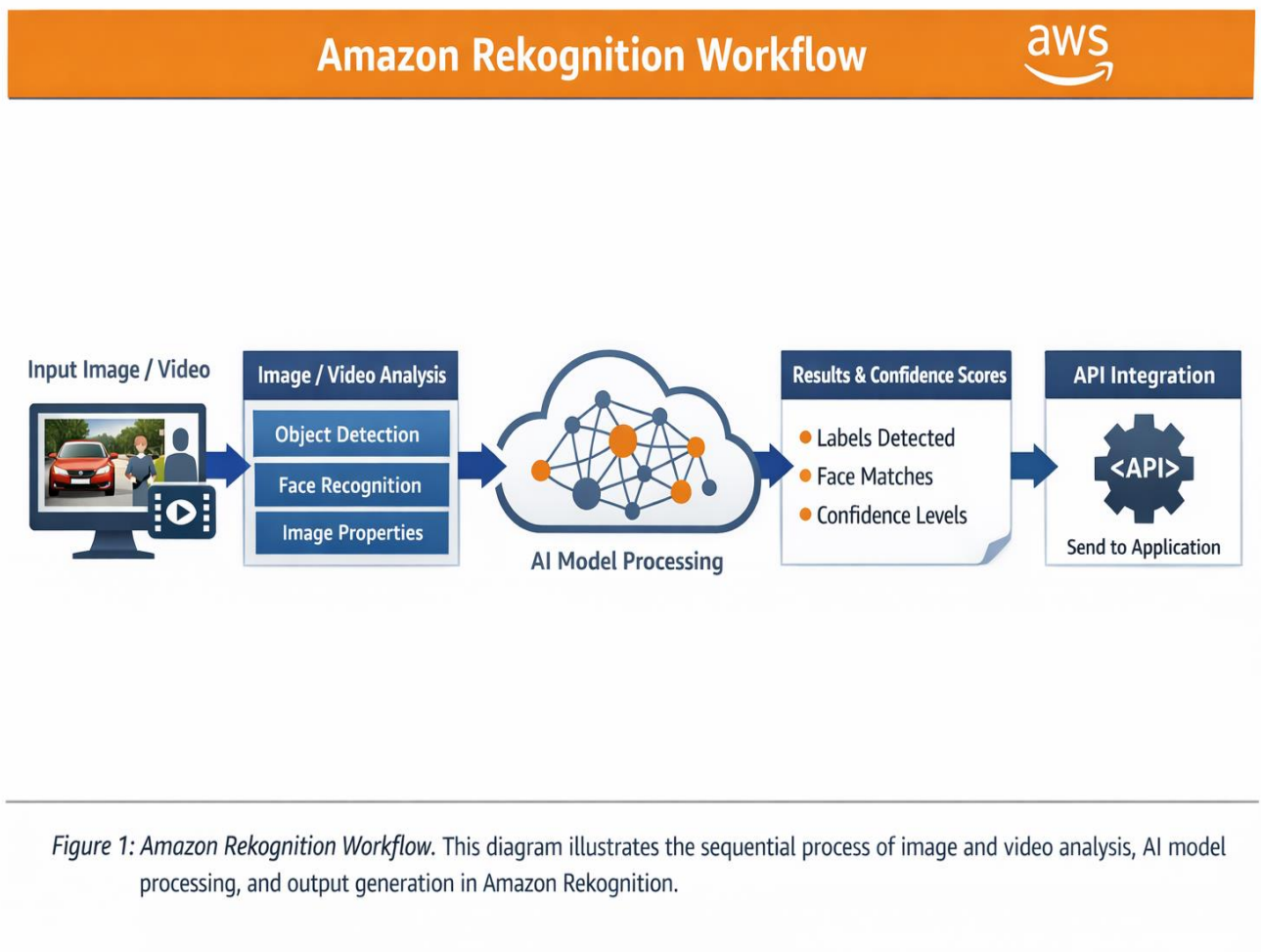


Figure 1: Amazon Rekognition Workflow. This diagram illustrates the sequential process of image and video analysis, AI model processing, and output generation in Amazon Rekognition.

Google AI Studio Exploration and Testing

Google AI Studio provided a markedly different interaction model centered on generative and multimodal capabilities.

User

Create an image of a kangaroo fighting a lion

Model

Here's that image for you!



 Google AI models may make mistakes, so double-check outputs.

In the first test, I prompted the system to generate an image of a kangaroo fighting a lion. The output was visually coherent, creative, and detailed. This demonstrated strong generative modeling but also highlighted that outputs are inherently probabilistic rather than verifiable.

User

Create an image of a man happily thinking about his true love.

Model



 Google AI models may make mistakes, so double-check outputs.

In the second test, I prompted the model to generate an image of a man happily thinking of his lover. The system successfully conveyed emotion through facial expression and contextual detail such as clothing. The image on the clothing reinforces the man is thinking about his lover. This reflects strong semantic modeling of emotional concepts.

User



Based on the image, here is a detailed description:

Subject and Action

The image is a composite photo showing the same man cloned three times in a living room setting, depicting him engaged in different technological activities simultaneously. The man has light skin, dark hair, and is wearing a dark blue baseball cap with a gold "GT" (Georgia Tech) logo, a black zippered fleece jacket, dark pants, and socks.

- **Left Figure:** Sitting on the far left of the couch, he is looking down at a smartphone held in both hands. A black cat is sleeping soundly on the top of the couch cushion directly behind his head.
- **Center Figure:** Sitting in the corner section of the sofa, he has a small, folding wooden tray table in front of him. A silver laptop is open on the table, and he is typing or looking at the screen.
- **Right Figure:** Sitting on the far right end of the sofa, he is holding a tablet in a dark case. His legs are stretched out on the sofa cushion. There is a noticeable digital editing artifact here, as his legs and lower body appear semi-transparent or "ghostly," revealing the leather couch beneath them.

Setting and Background

The scene takes place in a domestic living room with beige walls and reddish-brown wood laminate flooring.

- **Furniture:** The main piece of furniture is a large, brown leather L-shaped sectional sofa. It appears plush and comfortable. To the bottom right, there is a woven wicker storage cube or ottoman with a black remote control resting on top.
- **Decor:** On the wall to the left, behind the first figure, hangs a round decorative mirror with a frame that resembles a gear or sunburst design. To the right, a tall floor lamp with a dark base and a beige bell-shaped shade is turned on, casting a warm yellow light.
- **Details:** Several black power cords and a power brick are visible on the floor to the left, snaking towards the wall. White crown molding lines the top of the walls.

Start typing a prompt, use option + enter to append

In the third test, I uploaded an image and asked the system to describe it. The response was detailed and context rich. I think it exceeded the descriptive depth provided by Rekognition's label based outputs. This illustrates a key epistemic difference. Google AI Studio produces plausible narratives, while Rekognition produces discrete detections.

I did experience minor difficulty when attempting to create a video in Google AI Studio. This appeared to be usability friction rather than a fundamental limitation. It reinforced that advanced features may require additional familiarity and setup.

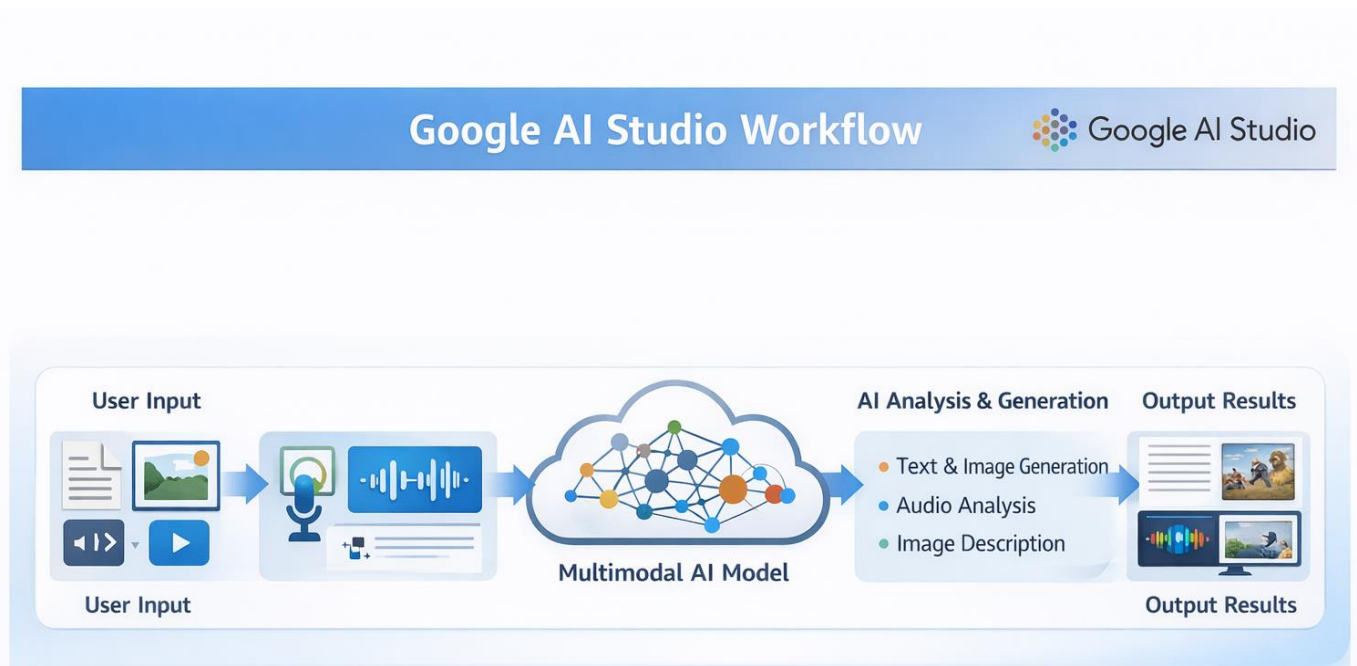


Figure 2: Google AI Studio Workflow. This diagram illustrates the process of generating and analyzing text, images and audio through a unified multimodal AI model in Google AI Studio.

Implementation Comparison and Decision Tradeoffs

When comparing Amazon Rekognition and Google AI Studio, the most significant difference lies in how each platform conceptualizes and communicates AI outputs. Amazon Rekognition is designed around deterministic computer vision tasks. It produces structured results such as labels, bounding boxes, face matches, and confidence scores. They are intended to support automated decision pipelines. This design makes Rekognition well suited for environments where predictability, repeatability, and auditability are critical. During my testing, the consistency of Rekognition's output format made it easier to imagine integration into larger systems. This could be useful in situations such as monitoring dashboards or evidence review workflows. However, this same structure can also limit flexibility because the system provides little interpretive context beyond what is explicitly detected.

Google AI Studio operates under a very different paradigm. Rather than focusing on detection and classification, it emphasizes generative and interpretive outputs across multiple modalities. When describing or generating images, the system produces rich narrative explanations rather than discrete labels. This makes the platform particularly effective for exploratory tasks, prototyping, and creative or

descriptive applications. In my testing, Google AI Studio demonstrated a strong ability to synthesize visual information into human readable explanations. This felt closer to natural reasoning. At the same time, the probabilistic nature of these outputs introduces variability that would be difficult to manage in regulated or high accountability environments. Unlike Rekognition, Google AI Studio does not provide confidence scores in a structured way. This makes it harder to assess reliability on a per output basis.

From a decision making perspective, the tradeoff between these platforms becomes clearer when considering risk tolerance and domain constraints. Amazon Rekognition favors precision and control at the cost of contextual richness. Google AI Studio favors flexibility and expressiveness at the cost of determinism. In healthcare and law enforcement contexts, where errors can have serious consequences, Rekognition's structured outputs and confidence signaling may be preferable, even if the system occasionally misses contextual details. Google AI Studio may be better suited for supportive roles. Google AI studio could be used for documentation assistance, training simulations, or exploratory analysis. This is because Google AI studios creativity and interpretive depth add value without directly driving decisions. This comparison reinforced for me that platform selection is less about which system is more advanced and more about which system aligns with the operational, ethical, and governance requirements of a given project.

Pricing further differentiates the two platforms in ways that affect implementation planning. Amazon Rekognition uses a clear per image and per request pricing model. This makes costs relatively predictable and easier to estimate during system design. This structure supports controlled scaling but can become expensive for high volume image or video analysis workloads. Google AI Studio pricing is tied to model usage and interaction volume. This offers flexibility for experimentation but makes long term cost forecasting more difficult. From an implementation standpoint, Rekognition favors stable production budgeting, while Google AI Studio favors iterative exploration where usage patterns may change frequently.

Platform Comparison: Amazon Rekognition vs Google AI Studio

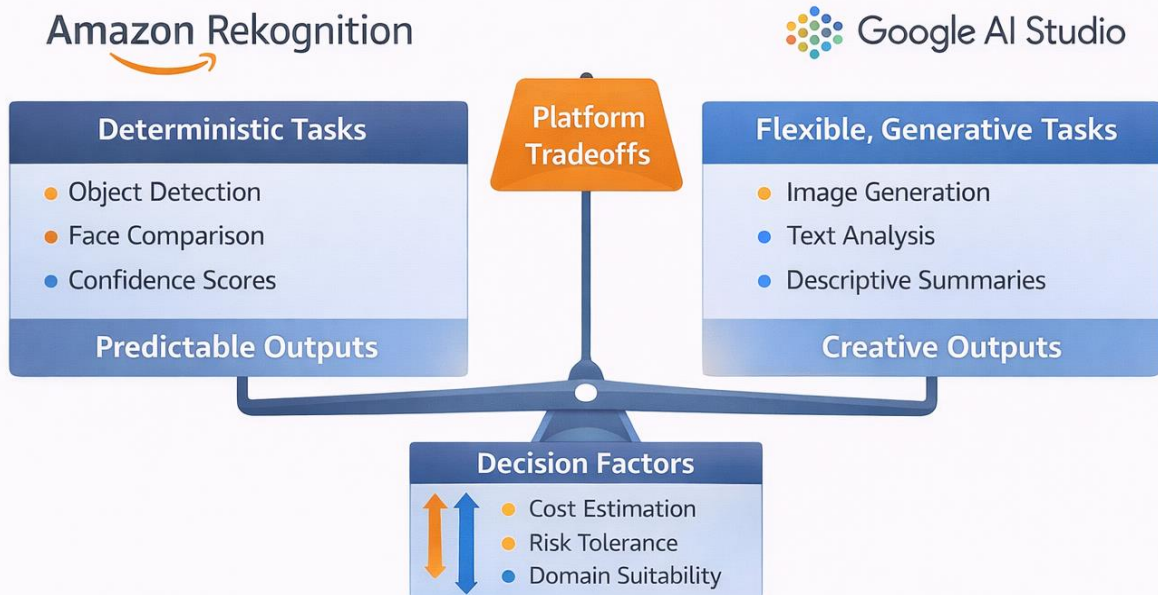


Figure 3: Platform Comparison. This diagram highlights the decision tradeoffs between deterministic tasks in Amazon Rekognition and generative tasks in Google AI Studio.

Domain Adaptation: Healthcare and Law Enforcement

In healthcare, Rekognition could support imaging triage or patient monitoring. This is feasible if confidence thresholds and false negative risks are clearly understood. False negatives are especially critical in healthcare because missed detections can delay care. Google AI Studio may be better suited for clinical documentation or patient education, where descriptive richness matters more than deterministic accuracy.

In law enforcement, Rekognition's face comparison and object detection capabilities may support evidence review. Though, inconsistent detection performance reinforces the need for human oversight. Google AI Studio could assist with report summarization or training simulations, though it is not appropriate for real time identification.

Ethical and Societal Considerations

Automation optimism remains high. Gartner reports that "80 percent of executives believe automation can be applied to any business decision" (Gartner, 2022). This belief increases the risk that AI outputs are treated as authoritative without sufficient scrutiny.

Cloud data governance also matters. ISACA warns that “cross border cloud storage introduces governance and compliance risks that organizations often underestimate” (ISACA, 2024). These risks are amplified in sensitive domains such as healthcare and law enforcement.

A third ethical consideration raised by the image containing the painting involves questions of intellectual property and ownership in AI based image analysis. When Amazon Rekognition labeled the objects within the painting but ignored the surrounding real world context, it raised questions about how AI systems interpret creative works versus physical environments. From an intellectual property perspective, analyzing and labeling artwork can blur boundaries between observation and derivative use. This is especially true when the artwork itself is protected by copyright. While Rekognition is designed to detect visual features rather than reproduce or reinterpret artistic meaning, the act of algorithmically classifying elements within a painting still treats creative expression as raw data. This highlights a broader ethical tension in AI systems that process images containing copyrighted or personally owned artwork. This is where ownership rights, consent, and appropriate use are often undefined or assumed rather than explicitly addressed.

Reflection and Lessons Learned

This comparison reinforced that AI capability does not equal AI suitability. Amazon Rekognition felt stable and predictable, but limited in contextual awareness. Google AI Studio felt powerful and expressive, but less constrained. If I were deploying either platform professionally, I would prioritize understanding failure cases, confidence interpretation, and domain alignment before scaling use.

Conclusion and Recommendations

Amazon Rekognition is better suited for structured, regulated environments that require predictable outputs and integration stability. Google AI Studio is better suited for exploratory, generative, and descriptive applications. Platform selection should be driven by domain risk tolerance rather than novelty.

As the NIST AI Risk Management Framework states, “managing AI risks requires ongoing assessment across the entire system lifecycle” (Tabassi, 2023). That principle aligns directly with my experience in this comparison.

References

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